ML

06 November 2025 00:33

1. Introduction to Networking

What is a Network?

A **network** is a collection of interconnected devices, like computers, servers, and other hardware, that communicate and share resources (such as files, internet access, and printers). Networks enable efficient information transfer and collaboration between devices and users.

- **Purpose of Networking**:
    - **Resource Sharing**: Allows multiple users to share devices (e.g., printers) and files.
    - **Communication**: Enables communication via emails, messaging, and video calls.
    - **Data Sharing**: Allows data to be accessed by different users, making it easier to collaborate.
    - **Centralized Data Management**: Simplifies data backups, management, and security.

Types of Networks

Networking types are classified by the size, structure, and geographic reach:

1. **Local Area Network (LAN)**:
    - **Scope**: Covers a small area, like an office or a building.
    - **Purpose**: Connects devices in a single location.
    - **Speed**: High data transfer rates (up to 10 Gbps).
    - **Example**: Office or home network.
2. **Wide Area Network (WAN)**:
    - **Scope**: Covers a large geographic area, such as cities, countries, or continents.
    - **Purpose**: Connects multiple LANs over large distances.
    - **Speed**: Slower than LANs (typically up to 200 Mbps).
    - **Example**: The internet is a global WAN.
3. **Metropolitan Area Network (MAN)**:
    - **Scope**: Covers a city or a large campus.
    - **Purpose**: Connects multiple LANs within a specific area.
    - **Example**: A city-wide network provided by an ISP.
4. **Personal Area Network (PAN)**:
    - **Scope**: Covers a very small area, typically a few meters.
    - **Purpose**: Connects personal devices for individual use.
    - **Example**: Bluetooth connections between a phone and wireless earbuds.

Network Topologies

Network topology describes how devices in a network are arranged and connected to each other. Different topologies have unique benefits and limitations.

1. **Bus Topology**:
    - **Structure**: All devices are connected to a single central cable, called a bus.
    - **Advantages**: Simple, easy to install, and low cost.
    - **Disadvantages**: If the main cable fails, the entire network goes down.
2. **Star Topology**:
    - **Structure**: All devices connect to a central hub or switch.
    - **Advantages**: If one device fails, it does not affect the others.
    - **Disadvantages**: If the central hub fails, the network is disrupted.
3. **Ring Topology**:
    - **Structure**: Devices are connected in a circular loop, with each device connected to two others.
    - **Advantages**: Simple data transfer path.
    - **Disadvantages**: Failure in any one connection can disrupt the entire network.
4. **Mesh Topology**:
    - **Structure**: Every device is connected to every other device.
    - **Advantages**: Highly reliable; one failure does not disrupt the network.
    - **Disadvantages**: Expensive due to the high number of connections required.
5. **Hybrid Topology**:
    - **Structure**: Combination of two or more topologies (e.g., Star-Ring or Star-Bus).
    - **Advantages**: Flexible, can be designed to meet specific needs.
    - **Disadvantages**: Complex setup and troubleshooting.

2. Network Architecture Models / Network Models

OSI and TCP layers are types of network architecture models or network model. These models provide structured frameworks that standardize how data is transmitted over a network.

Each model organizes communication tasks into layers, which define specific responsibilities, such as data encapsulation, transmission, and routing, making networking processes more efficient and modular. By dividing tasks into layers, these models ensure that devices and applications can communicate reliably and consistently, regardless of hardware or software differences.

OSI Model

The **OSI Model** is a framework that breaks down how data travels over a network into **seven layers**. It helps standardize communication between devices, so computers, phones, and other devices can talk to each other, even if they're made by different manufacturers. Created by the International Organization for Standardization (ISO), it divides the networking process into **seven layers**—each with specific roles to facilitate data transmission across diverse networks.

Why is the OSI Model Used?

The OSI model:

1. **Organizes**: It organizes network functions into layers so that each layer only needs to focus on one part of the communication process.
2. **Helps Troubleshooting**: If something goes wrong, we can pinpoint which layer has the issue, making it easier to fix.
3. **Ensures Compatibility**: Since each layer is standardized, different devices and technologies can work together smoothly.

OSI Model Layers in Simple Terms

Think of the OSI model as a **postal service** that delivers a letter (your data) to a friend. Each layer is like a step in this postal process:

1. **Physical Layer**: Think of it as the **road** or **mailbox**. This is the physical path the letter takes (wires, cables, etc.).
2. **Data Link Layer**: This layer is like the **address label** on the envelope. It ensures the letter can reach your friend's local post office correctly.
3. **Network Layer**: This layer is like the **postal system** that routes the letter between cities, finding the best path.
4. **Transport Layer**: This layer ensures the **letter gets delivered fully** (in case some pages are missing, they'll resend it).
5. **Session Layer**: Think of this as the **appointment** you set to talk with your friend so you're both available. It establishes a connection.
6. **Presentation Layer**: This layer is like **translating** the letter into a language your friend understands.
7. **Application Layer**: This is **your friend opening the letter** and reading it. It's the final step that lets applications use the data.

Example: Loading a Webpage

When you type a website URL in a browser:

1. **Application Layer**: The browser sends your request to access the webpage.
2. **Presentation Layer**: Formats or encrypts the data, so it's ready for transfer.
3. **Session Layer**: Sets up a session between your browser and the website.
4. **Transport Layer**: Breaks down your request into smaller pieces (segments) to send reliably.
5. **Network Layer**: Directs each piece (now packets) through routers to find the website's server.
6. **Data Link Layer**: Adds a physical address to send the packet over the internet.
7. **Physical Layer**: Converts your data into signals sent via cables or Wi-Fi.

When the website responds, the process reverses, and the webpage loads on your screen!

Layers of OSI Models

Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer

1. Physical Layer (Layer 1)

- **Function**: Deals with the physical transmission of raw bitstreams over a physical medium.
- **Responsibilities**:
    - Converting data to electrical, radio, or optical signals.
    - Managing hardware components like cables, switches, and network cards.
    - Defining physical characteristics (cable types, signal types, bit rate).
- **Examples**: Ethernet cables, fiber optics, Wi-Fi signals.
- **Protocols/Technologies**: Ethernet (physical components), USB, Bluetooth.

2. Data Link Layer (Layer 2)

- **Function**: Manages node-to-node data transfer and error detection on the same physical network.
- **Responsibilities**:
    - Organizes data into frames for transmission.
    - Provides MAC (Media Access Control) addresses for devices.
    - Detects and sometimes corrects errors that may occur at the Physical Layer.
- **Sub-layers**:
    - **MAC (Media Access Control)**: Manages access to the physical transmission medium.
    - **LLC (Logical Link Control)**: Ensures flow and error control over the network.
- **Examples**: Ethernet and Wi-Fi frames.
- **Protocols/Technologies**: Ethernet, Wi-Fi (IEEE 802.11), PPP (Point-to-Point Protocol), ARP (Address Resolution Protocol).

3. Network Layer (Layer 3)

- **Function**: Routes packets across networks and provides logical addressing (IP addresses).
- **Responsibilities**:
    - Determines the best path for data to reach its destination (routing).
    - Assigns IP addresses to devices and handles data packet forwarding.
    - Supports internetworking (connecting different networks).
- **Examples**: Routing packets from one router to another across different networks.
- **Protocols/Technologies**: IP (Internet Protocol), ICMP (Internet Control Message Protocol), IGMP (Internet Group Management Protocol).

4. Transport Layer (Layer 4)

- **Function**: Provides end-to-end communication, error checking, and data flow control.
- **Responsibilities**:
    - Ensures reliable data transfer between devices.
    - Breaks large data into smaller segments for easy transmission.
    - Reassembles segments at the destination and ensures they arrive correctly.
- **Examples**: Ensuring complete and correct file transfer during downloads.
- **Protocols/Technologies**:
    - **TCP (Transmission Control Protocol)**: Connection-oriented protocol for reliable data transfer.
    - **UDP (User Datagram Protocol)**: Connectionless protocol for faster, less reliable transmission.

5. Session Layer (Layer 5)

- **Function**: Manages sessions between applications, controlling dialog and data exchange.
- **Responsibilities**:
    - Establishes, maintains, and terminates connections (sessions) between devices.
    - Synchronizes communication and manages data exchange timing.
    - Controls session checkpoints and recovery in case of interruptions.
- **Examples**: Keeping a video call active and synchronized between two devices.
- **Protocols/Technologies**: NetBIOS, PPTP (Point-to-Point Tunneling